Mobile Programming (3 Cr.)

Course Code: CACS351

Year/Semester: III/VI

Class Load: 6Hrs. /Week (Theory: 3Hrs, Practical 3Hrs.)

Syllabus Discussion

Objectives

"Develop Mobile Application to solve real world problem using current mobile technology."

Internal Marks Distribution Categories

1. Laboratory Works
2. Presentations
3. Assignments
4. Class performance and attendance

Lets Start Learning To Develop Mobile App…

Unit-1: Introduction to Mobile and Mobile Programming

1.1 Mobile Device

- 1.1.1 Features
- 1.1.2 Categories,
- 1.1.3 History
- 1.1.4 Brands
- 1.1.5 Models and
- 1.1.6 Platforms

1.2 Introduction to Mobile Programming.

1.1 Mobile Device

- Mobile devices are portable or handheld electronic devices designed for communication, entertainment, and productivity purposes. These devices are small and lightweight, making them easy to carry around and use on-the-go.
- The revolutionized technology have changed the way we communicate, access information, and perform tasks on-the-go, and they continue to evolve with new technologies and features.
- Examples: Smartphones, Tablets, E-Readers and Smart Watches.

1.1 Mobile Device Features

1. Communication: Mobile devices allow users to make phone calls, send text messages, and use messaging apps to communicate with others.
2. Portable: Mobile devices are designed to be portable, allowing users to easily carry them around and use them on-the-go.
3. Battery: Mobile devices are powered by batteries that can be recharged and offer varying levels of battery life, depending on usage.
4. Storage: Mobile devices come with internal storage and often allow for expandable storage through the use of memory cards, enabling users to store a wide range of content such as photos, videos, and music.

1.1 Mobile Device Features [Contd…]

5. Wireless Connectivity: Mobile devices can connect to the internet and other devices using wireless technologies such as Wi-Fi, Bluetooth, and cellular data.
6. Mobile Applications: Mobile devices have access to app stores where users can download and install mobile applications for a wide range of purposes, including productivity, entertainment, education, and social media.
7. Personalization: Mobile devices offer various customization options such as wallpapers, themes, and app layouts, allowing users to personalize their device to their liking.

Other features: Internet Browsing, Camera, Sensors, GPS and so on.

1.1 Mobile Device Categories

Mobile devices can be categorized based on their size, speed, and functionality. Here are some of the main categories:

1. Feature Phones: (Bar Phones)
   - Basic mobile phones that primarily make calls and send texts.
   - Typically have limited functionality, small screens, and physical keypads.
   - Examples: Nokia 3310, Samsung Guru Music 2, Alcatel 10.66 etc.
2. Smartphones:
   - Advanced mobile phones that offer a range of features and functionality.
   - Typically have larger screens, touch interfaces, and access to app stores.
   - Examples: iPhone 13, Samsung Galaxy S21, Google Pixel 6 and so on.
3. Phablets:
   - Large-screen smartphones that blur the line between phones and tablets.
   - Typically have screens between 5.5 and 7 inches.
   - Examples: Samsung Galaxy Note20, iPhone 12 Pro Max, Google Pixel 6 Pro and so on.

1.1 Mobile Device Categories

4. Tablets:
   - Mobile devices with larger screens than smartphones and more powerful hardware.
   - Typically used for browsing, multimedia, and productivity tasks.
   - Examples: iPad Pro, Samsung Galaxy Tab S7, Microsoft Surface Pro 7 and so on.
5. Wearable's:
   - Mobile devices that are worn on the body, such as smart watches and fitness trackers.
   - Typically offer features such as health monitoring, notifications, and voice control.
   - Examples: Apple Watch Series 7, Samsung Galaxy Watch4, Fitbit Charge 5 etc.
6. Hybrid Devices:
   - Mobile devices that combine features of multiple categories, such as tablet-laptop hybrids or smartphone-tablet hybrids.
   - Typically offer flexibility and versatility in terms of how the device can be used.
   - Examples: Microsoft Surface Duo, Samsung Galaxy Fold3 and iPad Mini etc.

1.1 Mobile Device History

Years	Milestones
1973	First mobile phone call made by Martin Cooper
1983	Motorola DynaTAC 8000X introduced as first mobile phone
1992	IBM Simon introduced as the first smartphone
1996	Nokia 9000 Communicator introduced as a smartphone
2000	Ericsson R380 introduced as the first touchscreen phone
2002	BlackBerry introduced as a business-oriented smartphone
2007	Apple iPhone introduced, revolutionizing the smartphone
2008	First Android OS released by Google (September 23)
2009	Android 1.5 Cupcake released (April 27)
2010	Introduction of the first tablets, such as iPad
2010	iOS 4 released, introducing multitasking and FaceTime (June 21)

1.1 Mobile Device History

Years	Milestones
2011	First 4G LTE networks launched
2012	Android 4.1 Jelly Bean released, introducing Google Now (July 9)
2013	iOS 7 released, introducing flat design and Control Center (September 18)
2014	Introduction of the first smartwatches, such as Apple Watch
2014	Android 5.0 Lollipop released, introducing Material Design (November 12)
2019	First 5G networks launched
2019	iOS 13 released, introducing Dark Mode and Sign in with Apple (September 19)
Present	Continued development of mobile devices and operating systems with increasing power, features, and capabilities

1.1 Mobile Device Brands

There are many brands of mobile devices available in the market, each with its own unique features, designs, and price points. Here are some of the most popular mobile device brands:

1. Apple: Apple is known for its iPhone series of smartphones, as well as its iPad and Apple Watch devices.
2. Samsung: Samsung offers a range of mobile devices, including the Galaxy S and Galaxy Note series of smartphones, as well as its Galaxy Tab and Galaxy Watch devices.
3. Huawei: Huawei offers a range of smartphones, including the Mate and P series, as well as its MediaPad tablets and Watch GT smartwatches.
4. Xiaomi: Xiaomi is a Chinese brand that offers affordable smartphones, such as its Redmi and Mi series, as well as other mobile devices, including the Mi Pad tablet and Mi Band fitness tracker.

1.1 Mobile Device Brands [Contd…]

5. Google: Google offers the Pixel series of smartphones, as well as its Pixelbook laptop and Google Nest smart home devices.
6. OnePlus: OnePlus is a brand that focuses on high-end smartphones, such as its OnePlus series, known for their powerful performance and sleek designs.
7. LG: LG offers a range of mobile devices, including the G and V series of smartphones, as well as its G Pad tablets and LG Watch smartwatches.
8. Motorola: Motorola offers a range of smartphones, including its Moto G and Moto E series, as well as its Moto 360 smartwatches.

Other Brands include: Sony Ericsson, BlackBerry, Nokia and so on.

1.1 Mobile Device Models and Platforms

Mobile Devices	Models	Platforms
iOS	iPhone, iPad, and iPod Touch	Run on Apple’s iOS OS
Android	Samsung Galaxy, Google Pixel, OnePlus	Run on Google's Android OS
Windows	Microsoft Surface and Nokia Lumia	Run on Microsoft's Windows OS
BlackBerry Devices	BlackBerry Key2 and BlackBerry Motion	Run on BlackBerry's proprietary OS

1.2 Introduction to Mobile Programming

- Mobile programming refers to the process of developing applications or software that run on mobile devices such as smartphones and tablets.
- Mobile programming requires knowledge of specific programming languages, tools, and frameworks that are used to create mobile apps.
- There are several programming languages that can be used for mobile development, including:

1. Java: Java is a popular language used for Android app development.
2. Swift: Swift is a programming language used for developing apps for Apple's iOS and macOS platforms.

1.2 Introduction to Mobile Programming [Contd…]

3. Kotlin: Kotlin is a newer programming language used for Android app development, and it is now the preferred language for Android development by Google.

4. Objective-C:

Objective-C is a language used for developing apps for Apple's iOS and macOS platforms, although it has been largely replaced by Swift in recent years.

5. C#:

C# is a language used for developing apps for Microsoft's Windows Phone platform, although it is not as widely used as Java or Swift.

Mobile Application Development Approaches

1. Native app development:
   - Building apps for a specific platforms such as android or ios.
   - Best performance and user experience but requires more time and resources
   - iOS: Swift or Objective-C using Xcode IDE
   - Android: Java or Kotlin using Android Studio IDE
2. Cross-platform app development:
   - Building apps that can run on multiple platforms using a single codebase.
   - Provide native-like performance and user experience, but with less development time and resources
   - Example: React Native and Flutter

Mobile Application Development Approaches

4. Hybrid app development:
   - Building apps using web technologies such as HTML, CSS, and JavaScript, and wrapping them in a native container to run as a mobile app.
   - Example: Ionic (based on Angular), Apache Cordova
5. Low-code/no-code app development:
   - Allow developers to create mobile apps using drag-and-drop interfaces and pre-built components, without needing to write code.
   - Example: Bubble, Power Apps

Mobile Application Development Approaches

1. Planning
   - Define app goals and objectives
   - Identify target audience
   - Determine app features and requirements
   - Create project plan
2. Design
   - Design app UI/UX
   - Create wireframes and prototypes
   - Define app architecture and database structure
3. Development
   - Write code for the app
   - Integrate app features and APIs
   - Test app functionality

Mobile Application Development Approaches

4. Testing
   - Conduct manual testing by development team
   - Conduct user testing to gather feedback on app usability and functionality
   - Fix bugs and errors
5. Deployment
   - Submit app to app stores for distribution
   - Configure pricing and distribution settings
   - Market the app to users
6. Maintenance
   - Update app to work with new operating systems and devices
   - Fix bugs and errors
   - Add new features and functionality as needed

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END OF UNIT-1

Summary:

Mobile Devices:

- Refers to portable computing devices that are designed to be carried around.
- Includes smartphones, tablets, wearable's, and other handheld devices.

Features of Mobile Devices:

- Portability, camera, internet connectivity, mobile applications, and more.

Categories of Mobile Devices:

- Based on size, speed, and functionality.
- Categories include feature phones, smartphones, phablets, tablets, wearables, and hybrid devices.

Mobile Devices Brands:

- Examples include Apple, Samsung, Google, Huawei, Xiaomi, and many others.

END OF UNIT-1

Mobile Devices Models and Platforms:

- Examples of mobile devices model include the iPhone, Samsung Galaxy, Microsoft Surface, and BlackBerry Key2.
- Mobile devices run on different platforms such as iOS, Android, Windows, BlackBerry etc.

Mobile Application Development:

- Involves designing, building, testing, and deploying mobile applications.
- Development approaches include native, hybrid, and web apps.

Mobile Application Development Cycle:

- Involves planning, designing, developing, testing, deploying, and maintaining mobile apps.
- A continuous iterative process that requires ongoing improvement and iteration.

END OF UNIT-1

Mobile Application Development Approaches:

- Native apps: developed for a specific platform (e.g. iOS, Android) using platform-specific languages and tools.
- Hybrid apps: developed using web technologies (HTML, CSS, JavaScript) and wrapped in a native app shell to run on multiple platforms.
- Web apps: run in mobile web browsers and are not installed on the device.

NEXT

Unit 2 : Introduction of Android Programming

Prerequisites:

- Watch Environmental Setup for Android videos on YouTube and setup Android Studio on your laptop.

“ **Have a Good Day** ”

Thank You